Alpha decay 2

Let ν be the rate of tunneling attempts. Decay rate λ is ν times tunneling probability.

$$\lambda = \nu e^{-2\gamma}$$

Gamow theorized that ν is proportional to the inverse of r_1 because λ increases as atomic weight decreases. By dimensionality the numerator has to be a velocity v.

$$\nu = \frac{v}{2r_1}$$

Velocity v is unknown so let our hypothesis be that v is proportional to $\sqrt{E/m}$.

$$\lambda = \frac{\beta\sqrt{E/m}}{2r_1} e^{-2\gamma}$$

Coefficient β is estimated to be

$$\beta = 843$$

Lifetime τ is the inverse of λ.

$$\tau = \frac{1}{\lambda}$$

Half-life $\tau_{1/2}$ is a fraction of τ.

$$\tau_{1/2} = \frac{\log 2}{\lambda} = \log 2 \frac{2r_1}{\beta\sqrt{E/m}} e^{2\gamma}$$

Hence

$$\tau_{1/2} = \frac{2\log 2}{\beta} \frac{r_1}{\sqrt{E/m}} \exp\left(2C_1 \frac{Z}{\sqrt{E}} - 2C_2\sqrt{Zr_1}\right)$$

For the alpha decay process ${}^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th}$ we have

$$\begin{aligned}
A &= 238 \\
Z &= 90 \\
E &= 4.27\,\mathrm{MeV} \\
r_1 &= 7.44 \times 10^{-15}\,\mathrm{m}
\end{aligned}$$

Hence

$$\tau_{1/2} = 3.9 \times 10^{17}\,\mathrm{s} = 12 \times 10^{9}\,\mathrm{yr}$$

The actual half-life of ${}^{238}\mathrm{U}$ is

$$4.5 \times 10^{9}\,\mathrm{yr}$$

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